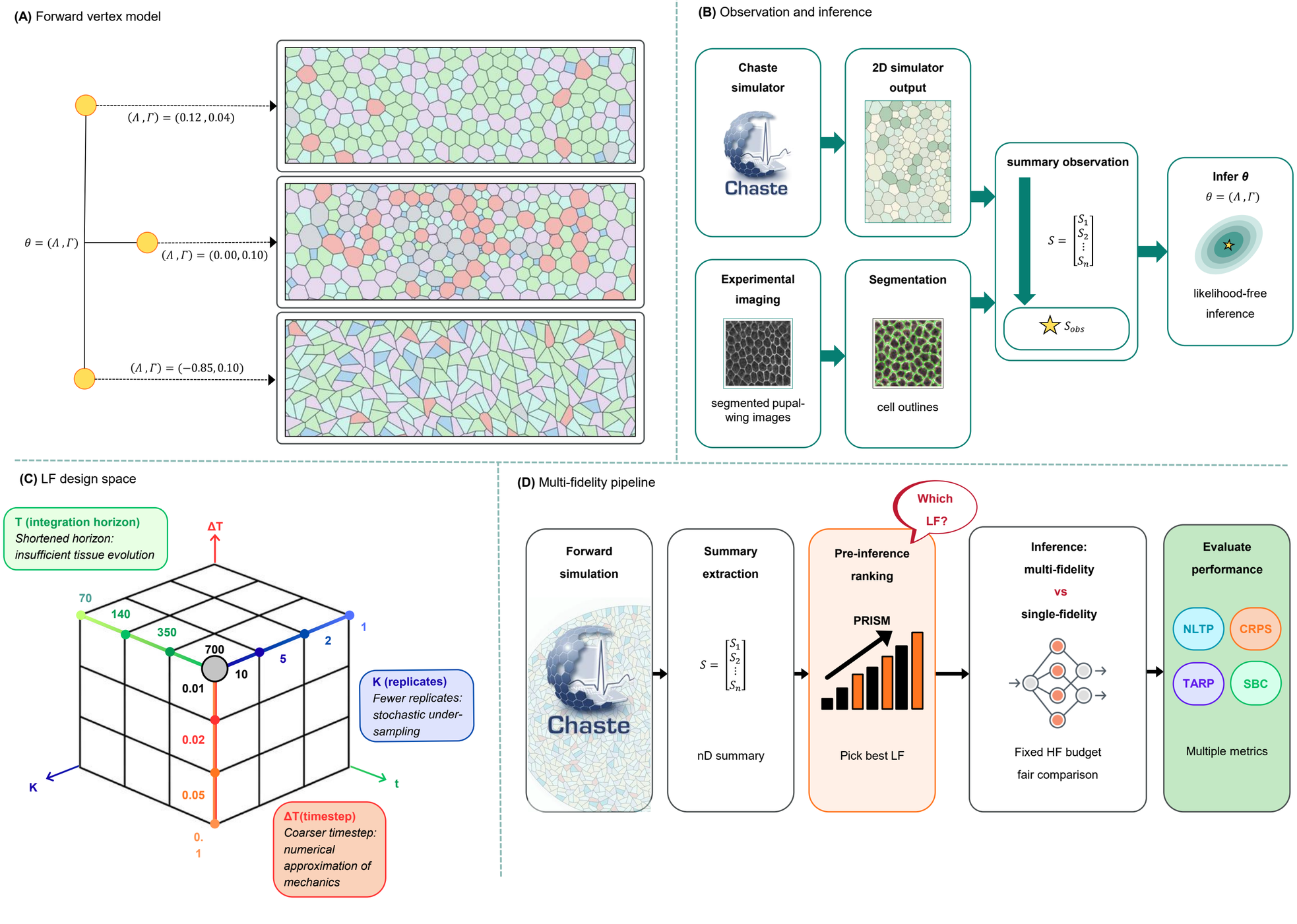

(A) Forward vertex model
$\theta = (\Lambda, \Gamma)$
$(\Lambda, \Gamma) = (0.12, 0.04)$
$(\Lambda, \Gamma) = (0.00, 0.10)$
$(\Lambda, \Gamma) = (-0.85, 0.10)$
(B) Observation and inference
Chaste simulator
Chaste
2D simulator output
Experimental imaging
segmented pupal-wing images
Segmentation
cell outlines
summary observation
$S = \begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_n \end{bmatrix}$
S_{obs}
Infer θ
$\theta = (\Lambda, \Gamma)$
likelihood-free inference
(C) LF design space
T (integration horizon)
Shortened horizon: insufficient tissue evolution
ΔT
70
140
350
700
10
5
2
1
0.01
0.02
0.05
0.1
K
t
K (replicates)
Fewer replicates: stochastic under-sampling
ΔT(timestep)
Coarser timestep: numerical approximation of mechanics
(D) Multi-fidelity pipeline
Forward simulation
Chaste
Summary extraction
$S = \begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_n \end{bmatrix}$
nD summary
Which LF?
Pre-inference ranking
PRISM
Pick best LF
Inference: multi-fidelity vs single-fidelity
Fixed HF budget fair comparison
Evaluate performance
NLTP
CRPS
TARP
SBC
Multiple metrics